RUBBERFORM®
RECYCLED PRODUCTS, LLC

75 Michigan Street
Lockport, NY 14094

Quality Management System "QMS"

1/4/2026

Version 26.0

26.0

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Quality Policy

RubberForm Recycled Products LLC. has developed and implemented a comprehensive **Quality Management System (QMS)** to standardize its business processes, enhance customer satisfaction, and improve overall operational performance. The primary objectives of the QMS are to ensure that RubberForm meets or exceeds customer requirements, complies with relevant regulatory standards, and continuously optimizes product quality and efficiency.

This manual details the structure of the QMS, specifying the roles, authorities, and responsibilities of personnel involved in its use. It also provides standardized procedures, guidelines, and references for all activities within the QMS, ensuring adherence to the required industry standards and regulatory compliance.

Internally, the manual functions as an operational guide to align employees with company goals, focusing on customer satisfaction, continuous process improvement, and empowering personnel to take initiative in maintaining product quality.

Statement of Authority

The Quality Program outlined in this manual serves as the definitive Governing document for the Engineering, Manufacturing, Inspection, Assembly, Packaging, and Shipping of RubberForm products.

The designated Quality System representatives, listed below, hold responsibility for the preparation, review, revision, and distribution of the Quality Manual, and for managing the administration and implementation of the entire Quality Program. These representatives are granted full authority within their respective areas to identify quality control issues, analyze root causes, and initiate corrective and preventive actions (CAPA) to ensure ongoing compliance with this manual and relevant industry standards.

Furthermore, these representatives possess the authority to halt operations when quality discrepancies arise and to mediate disputes between departments regarding quality-related issues. If a conflict arises between a Quality System representative and any other managerial or operational staff member, the issue will be escalated to executive management for final resolution, who will determine the company's disposition, without compromising the integrity of RubberForm's Quality System requirements.

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Quality System Representatives

The undersigned representatives of the Quality System agree to uphold the principles and guidelines of this Quality Manual with professionalism and expertise. They also commit to supporting executive leadership, adhering to the Quality Manual's standards, and ensuring the integrity of RubberForm's product quality and operations.

We recognize that our shared dedication to quality is the foundation of RubberForm's success. By embracing these standards, each of us contributes to creating products we can all take pride in, strengthening our reputation, and building trust with our customers. Together, we can achieve excellence, foster innovation, and ensure a safe, supportive workplace where everyone's contributions are valued and celebrated. Let this commitment inspire us to not only meet expectations but to exceed them in everything we do.

Position	**Name**	**Signature**	**Date**
President	Bill Robbins		
Operations	Jeff Robbins		
Finance	John Clouden		
Sales/Marketing	Jake Robbins		
Engineering	Joe Probst		
Production	Brian Hamp		
Maintenance	Anthony Cescon		
Quality	Anthony/Brian/Joe		

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Master Document List

Doc. ID	Doc. Name	Edit Type	Release Type	Owner	Approved By:	Level
P-01	Document Control	Word	PDF	Engineering	Admin	
P-02	Record Control	Word	PDF	Admin	Engineering	
P-03	Design Control	Word	PDF	Engineering	Sales	
P-04	Management Review	Word	PDF	Admin	Accounting	
P-05	Training	Word	PDF	Production	Engineering	
P-06	Production and Process Control	Word	PDF	Production	Admin	
P-07	Order Processing and Review	Word	PDF	Sales	Production	
P-08	New Product Development	Word	PDF	Engineering	Sales	
P-09	Purchasing	Word	PDF	Accounting	Admin	
P-10	Incoming Inspection	Word	PDF	Production	Admin	
P-11	In process Inspection	Word	PDF	Production	Engineering	
P-12	Customer Complaint	Word	PDF	Sales	Admin	
P-13	Non-Conformance	Word	PDF	Engineering	Admin	
P-14	Corrective/Preventive Action	Word	PDF	Engineering	Admin	
P-15	Drawing Control	Word	PDF	Engineering	Admin	
P-16	Internal Work Order	Word	PDF	Production	Engineering	
P-17	File Naming	Word	PDF	Admin	Engineering	
P-18	Work Instruction Creation	Word	PDF	Engineering	Production	
P-19						
P-20	General Maintenance	Word	PDF	Production	Admin	
F-00F	Form Template			Admin	Engineering	
F-00P	Procedure Template			Admin	Engineering	
F-01	Document Revision Request			Engineering	Admin	
F-02						
F-03						
F-04						
F-05						
F-08	New Product			Sales	Engineering	
F-09	Purchase Request			Production	Accounting	
F-10	Incoming Inspection			Production	Engineering	
F-10-1	In Process Inspection			Production	Admin	
F-12	Customer Complaint			Sales	Engineering	
F-13	Non-Conformance			Production	Admin	
F-14	Corrective / Preventive Action			Engineering	Admin	
F-16	Work Order Request			Production	Admin	
F-16-1	Maintenance Request			Production	Engineering	
F-16-2	Equipment Status			Engineering	Production	

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Master Document List (continued)

Doc ID	Doc. Name	Edit Type	Release Type	Owner	Approved By:	Level
WI-01	Updating QMS Manual					
WI-02	Training Assessment					
WI-03						
WI-04						
WI-05						
WI-06						
WI-07						
WI-05						
WI-06						
WI-07						
WI-31	Compression Mold Polishing					
WI-41	Hydraulic Cylinder Swap Press 3					
RD-01	Organizational Chart					
RD-02	Roles and Responsibilities					
RD-03	Training Matrix					
RD-04	Maintenance Meeting Agenda					
RD-05	Production Meeting Agenda					
RD-06	Engineering Meeting Agenda					
RD-07						

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Training Matrix

To be developed

Organizational Chart

Still need this.....

Procedures and Forms

1. Purpose

1.1. Clearly state the purpose of this procedure.

2. Scope

2.1. Define the boundaries of the procedure, including where it applies and any exclusions.

3. References

3.1. ISO 9001
3.2. GMP Guidelines
3.3. Company Quality Management System (QMS)

4. Definitions (Optional)

4.1. Include any terms or abbreviations used in the procedure.

5. Responsibilities

5.1. Identify roles and responsibilities.

6. Procedure

6.1. Provide step-by-step instructions

7. Supporting Documents

7.1. - Attach relevant templates, checklists, or forms.

8. Recordkeeping

8.1. Detail what records must be kept, how, and for how long.

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.
9.2. Revised documents require review and approval before implementation.
9.3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
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1. Purpose

1.1. The Document Control procedure ensures that all QMS documents within the company are properly created, reviewed, approved, distributed, and maintained to support consistency, compliance, and traceability.

2. Scope

2.1. This procedure applies to all controlled documents within the QMS, including but not limited to:
- 2.1.1. Policies, procedures, drawings, and work instructions.
- 2.1.2. Forms, templates, and records.
- 2.1.3. Manuals, guidelines, and specifications.
- 2.1.4. Internal and external documents relevant to compliance, quality, and operations.

2.2. This procedure covers the entire document lifecycle, including but not limited to:
- 2.2.1. Creation & approval – Establishing document formats, assigning responsibilities, and obtaining authorization.
- 2.2.2. Revision & version control – Managing changes, approvals, and tracking document history.
- 2.2.3. Distribution & Accessibility – Ensuring authorized personnel have access to the latest versions.
- 2.2.4. Storage & Retention – Defining secure storage and retention periods in accordance with legal and operational requirements.
- 2.2.5. Obsolescence & Disposal – Managing the removal and archiving of outdated or superseded documents.

2.3. This procedure applies to all personnel responsible for creating, modifying, reviewing, approving, storing, distributing, or using controlled documents.

3. References

3.1. ISO 9001
3.2. GMP Guidelines
3.3. Company Quality Management System (QMS)

4. Definitions (Optional)

4.1. Controlled Document: A document which cannot be changed without justification, review, and approval by two (2) or more management positions. Typically identified by a revision number or letter, applicable date, and name or signature of at least one person authorized to approve controlled documents.

4.2. Document Approval: Affirmation that a controlled document is approved, as amended, and may be used by company personnel. Typically indicated by at least one name or signature of a person authorized to approve documents. This shall be accompanied by a change form, or statement in the document explaining the change. A simple revision history on the document may be used if it describes or directs the reader/user to the details for the change.

4.3. Authorized Person: An employee of the company who may approve documents. Generally, a management level person in Engineering, Operations, Quality, Sales, etc. as applicable.

5. Responsibilities

5.1. Document owners: responsible for the creation, updates, and accuracy of controlled documents.
5.2. Revoewers & Approvers: Ensure that documents comply with company policies and regulatory requirements.
5.3. Document Control Administrator: Manages the document control system, maintains records, and enforces document control policies.
5.4. End Users: Ensure they are using the latest approved versions of documents.

6. Procedure

6.1. Document Creation & Approval.
- 6.1.1. Identify the need for a new document by reviewing operational or regulatory requirements.
- 6.1.2. Download and use an approved document template.
- 6.1.3. Assign a unique document identification number, revision number, and effective date. An electronic file name and/or path may be used provided that the document is uniquely identified.
- 6.1.4. Complete all necessary sections with clear, concise, and accurate information. See Document Change Request Form.
- 6.1.5. Submit the document to the designated Reviewer & Approver for validation.
- 6.1.6. Incorporate feedback, if any, and finalize the document.
- 6.1.7. Obtain final approval from an authorized individual before release.
- 6.1.8. Upload the approved document to the controlled document repository.
- 6.1.9. Notify all relevant personnel of the new document and provide access instructions.

6.2. Document Revision & Version control.
- 6.2.1. Identify the need for document modifications (e.g., regulatory updates, process changes, error corrections).
- 6.2.2. Retrieve the current version from the controlled document repository.
- 6.2.3. Edit the document while maintaining a revision history log.
- 6.2.4. Clearly document changes and provide a rationale for the revision. See Document Change Request Form.
- 6.2.5. Submit for review and approval following the same steps as the initial document creation process.
- 6.2.6. Archive the previous version and label it as “Obsolete”.
- 6.2.7. Update all relevant personnel about the new version.

6.3. Distribution & Accessibility.
- 6.3.1. Ensure that only the latest approved versions are accessible to users.
- 6.3.2. Store documents in a secure, controlled repository with appropriate user access permissions.
- 6.3.3. Communicate any updates via email or internal notification systems.

6.4. Storage & Retention
- 6.4.1. Securely store documents as per legal and regulatory retention requirements.
- 6.4.2. Define and document retention periods in a accordance with company policy.
- 6.4.3. Conduct periodic reviews of stored documents to ensure they remain relevant and compliant.

6.5. Obsolescence & Disposal.
- 6.5.1. Identify documents that are no longer in use or superseded by a newer version.
- 6.5.2. Move obsolete documents to an archived location within the repository.
- 6.5.3. For physical copies, securely shred or dispose of them according to company policies.
- 6.5.4. Maintain an Obsolete Document Log for audit tracking.

6.6. Compliance & Auditing.
   6.6.1. Conduct periodic internal audits to verify compliance with document control procedures.
   6.6.2. Review document management practices during QMS compliance assessments.
   6.6.3. Implement corrective actions if any document control deficiencies are identified.

7. Supporting Documents

7.1. Appendix A: Document Change Request Form.

8. Recordkeeping

8.1. Records will be stored electronically within the Quality Management System for a minimum of three (3) years.
8.2. Obsolete documents will be archived for historical reference.

9. Revision History

9.1. Any modifications must be documents with a revision history, including changes made and the reason for the update.
9.2. Revised documents require review and approval before implementation.
9.3. Old versions must be archived and marked as “Obsolete”.

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
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Appendix A

F-01 Document Revision Request JP

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1. Purpose

1.1. The purpose of this Control of Records Procedure is to establish a systematic approach for the identification, storage, protection, retrieval, retention, and disposal of records within the Quality Management System (QMS). This ensures compliance with regulatory requirements, data integrity, and accessibility of critical information.

2. Scope

2.1. This procedure applies to all records that are generated, maintained, and retained within the QMS, including but not limited to:

2.1.1. Quality records such as inspection reports, audit findings, and corrective actions.

2.1.2. Production and operational records.

2.1.3. Training records.

2.1.4. Customer complaints and feedback.

2.1.5. Legal and regulatory compliance records.

2.2. This procedure does not apply to informal communications such as personal notes, emails, or temporary working documents.

3. References

3.1. ISO 9001

3.2. GMP Guidelines

3.3. Company Quality Management System (QMS)

4. Definitions (Optional)

4.1. Record: Documented information that provides evidence of activities performed or results achieved.

4.2. Retention Period: The length of time a record must be maintained before disposal.

4.3. Archive: A secure location where obsolete or historical records are stored.

5. Responsibilities

5.1. Record Owners: Responsible for the creation, maintenance, and accuracy of records.

5.2. Department Heads: Ensure compliance with record control requirements within their respective areas.

5.3. Quality Manager: Oversees the implementation of the Control of Records procedure and ensures adherence to retention policies.

5.4. Document Control Administrator: Manages record storage, retrieval, and disposal processes.

6. Procedure

6.1. Record Identification & Creation

6.1.1. Identify records that require control based on regulatory, operational, and quality requirements.

6.1.2. Ensure all records are created using approved templates or formats.

6.1.3. Assign unique identifiers where applicable (e.g., serial numbers, document codes).

6.2. Storage & Protection

6.2.1. Store records in a designated secure location (physical or digital repository).

6.2.2. Implement access controls to prevent unauthorized modifications or deletions.

6.2.3. Protect records from damage, deterioration, or loss through environmental controls and security measures.

6.3. Retrieval & access

6.3.1. Ensure records are easily retrievable by authorized personnel when needed.

6.3.2. Implement an indexing or cataloging system for quick reference.

6.3.3. Define access levels to restrict unauthorized use of sensitive records.

6.4. Retention & Disposal

6.4.1. Define and document retention periods for different record types as per company policy and regulatory requirements.

6.4.2. Review records periodically to determine if they should be archived or disposed of.

6.4.3. Dispose of obsolete records through secure destruction methods (e.g., shredding, digital deletion) to prevent unauthorized access.

6.4.4. Maintain a Record Disposal Log for tracking purposes.

6.5. Compliance & Auditing

6.5.1. Conduct periodic internal audits to verify compliance with the Control of Records procedure.

6.5.2. Review recordkeeping practices during QMS compliance assessments.

6.5.3. Implement corrective actions if any deficiencies are identified in record management practices.

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7. Supporting Documents

7.1.

7.2.

7.3.

8. Recordkeeping

8.1. Records will be maintained as per the defined retention periods.

8.2. Obsolete records will be archived for historical reference when applicable.

8.3. Record disposal will be documented and reviewed periodically.

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.

9.2. Revised documents require review and approval before implementation.

9.3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
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1. Purpose

1.1. The purpose of this Design Control Procedure is to establish a structured approach for the planning, development, review, verification, validation, and approval of product designs within the Quality Management System (QMS). This ensures product safety, performance, and compliance with regulatory requirements.

2. Scope

2.1. This procedure applies to all design and development activities, including:

2.1.1. New product design and development

2.1.2. Modifications to existing designs

2.1.3. Process design changes that impact product quality

2.1.4. Design transfer to production

2.2. This procedure does not apply to non-product related improvements or minor process optimizations that do not affect product specifications.

3. References

3.1. ISO 9001:2015 Section 8.3 (Design and Development)

3.2. GMP Guidelines

3.3. Company Quality Management System (QMS)

4. Definitions (Optional)

4.1. Design Verification: Process of evaluating whether a design meets specified requirements.

4.2. Design Validation: Process of testing a design to ensure it meets user needs and intended use.

4.3. Design Transfer: Process of ensuring that design outputs are correctly transferred to production or implementation.

5. Responsibilities

5.1. Design Engineers: Responsible for developing and documenting design specifications and ensuring compliance with requirements.

5.2. Project Managers: Oversee design control activities and ensure milestones are met.

5.3. Quality Assurance Team: Conduct design reviews, verification, and validation to ensure regulatory and quality compliance.

5.4. Manufacturing Team: Support design transfer and provide input for manufacturability.

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6. Procedure

6.1. Design Planning

6.1.1. Define project scope, objectives, and regulatory requirements.

6.1.2. Develop a Design and Development Plan outlining responsibilities, timelines, and resources.

6.1.3. Obtain management approval before proceeding to the next phase.

6.2. Design Inputs

6.2.1. Identify and document design requirements, including customer needs, regulatory constraints, and functional specifications.

6.2.2. Ensure design inputs are clear, complete, and verifiable.

6.3. Design & Development

6.3.1. Develop preliminary designs and detailed specifications.

6.3.2. Create prototypes or models as needed.

6.3.3. Conduct internal design reviews to evaluate feasibility and adherence to requirements.

6.4. Design Verification

6.4.1. Conduct verification activities such as testing, inspections, and analysis to confirm design outputs meet input requirements.

6.4.2. Document verification results and resolve any identified issues before proceeding to validation.

6.5. Design Validation

6.5.1. Perform validation testing in real-world or simulated conditions.

6.5.2. Ensure the design meets user needs and intended operational conditions.

6.5.3. Document validation results and implement corrective actions if necessary.

6.6. Design Review & Approval

6.6.1. Hold formal design reviews at key milestones.

6.6.2. Approve final design after successful verification and validation.

6.6.3. Ensure all required documentation is completed and maintained.

6.7. Design Transfer

6.7.1. Ensure design outputs are accurately communicated to manufacturing or implementation teams.

6.7.2. Provide necessary training and support for production personnel.

6.7.3. Validate production processes to confirm they meet design requirements.

6.8. Design Changes & Control

6.8.1. Document and evaluate all proposed design changes.

6.8.2. Assess the impact of changes on product performance, regulatory compliance, and production processes.

6.8.3. Implement approved changes following the same verification and validation process.

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7. Supporting Documents

7.1. Document Control Procedure (P-01)
7.2. New Product Procedure (P-08)

8. Recordkeeping

8.1. Design documentation, including design inputs, outputs, verification, and validation records, must be retained for a minimum of 5 years.
8.2. Approved design changes must be documented and archived for future reference.
8.3. All design records must be securely stored and accessible for audits and regulatory inspections.

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.
9.2. Revised documents require review and approval before implementation.
9.3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
B				

26.0

1. Purpose

1.1. Ensure Top Management periodically evaluates QMS performance and drives continual improvement.

2. Scope

2.1. All QMS elements; focus on policy, objectives, audit results, customer feedback.

3. References

3.1. ISO 9001
3.2. GMP Guidelines
3.3. Company Quality Management System (QMS)

4. Definitions

4.1. Review Inputs: performance metrics, audit reports, nonconformances, feedback

5. Responsibilities

5.1. Quality Manager: schedule meeting, prepare agenda & inputs.
5.2. Top Management: attend, make decisions, assign actions.

6. Procedure

6.1. Prepare agenda.
6.2. Distribute to attendees.
6.3. Conduct review.
6.4. Record decisions & action items.
6.5. Assign owners & deadlines.
6.6. Follow up at next review.

7. Supporting Documents

7.1. Management Review Agenda template; Meeting Minutes form.

8. Recordkeeping

8.1. Records will be stored electronically in the Quality Management System for a minimum of 3 years.
8.2. Obsolete documents will be archived for historical reference.

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9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.

9.2. Revised documents require review and approval before implementation.

9.3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
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1. Purpose

1.1. Establish a structured approach for identifying training needs, delivering training, verifying effectiveness, and maintaining competency records to ensure work affecting quality is performed by competent personnel.

2. Scope

2.1. All positions, roles, and temporary assignments that impact product or service quality.

2.2. New hires, role changes, new equipment/processes, customer/regulatory requirements, and refresher cycles.

3. References

3.1. ISO 9001:2015 Section 7.2 (Competence)

3.2. GMP Guidelines

3.3. Company Quality Management System (QMS)

4. Definitions (Optional)

4.1. Competence: Demonstrated ability to apply knowledge and skills to achieve intended results.

4.2. OJT: On the job training with documented demonstration of skill.

4.3. Training Matrix: Role by skill map listing required qualifications and requalification intervals.

5. Responsibilities

5.1. Employees: Participate in training; demonstrate competence; follow procedures/WIs.

5.2. Department Managers: Identify training needs; assign trainers; verify effectiveness.

5.3. Quality: Own Training Matrix; monitor completion/effectiveness; report KPIs.

5.4. HR/Admin: Maintain records; manage expirations and requalification scheduling.

6. Procedure

6.1. Identification & Planning: Update the Training Matrix with method, criteria, due dates, and triggers (new hire, role change, new process/equipment, NCR/complaint, refresher).

6.2. Delivery: Conduct OJT/classroom/e learning/vendor training using current documents and job breakdowns.

6.3. Effectiveness Verification: Quiz ≥80%, practical demo via Job Qualification Checklist, first piece audit, or quality audit.

6.4. Records & Control: Complete Training Record including content, trainer/trainee, results, and evidence; file external certificates and track expirations.

6.5. Requalification & Awareness: Requalify before expiration; all personnel trained on Quality Policy, objectives, contributions to conformity, and consequences of nonconformance.

7. Supporting Documents

7.1. Training Record .

7.2. Training Matrix.

7.3. Job Qualification Checklist.

8. Recordkeeping

8.1. All training records, matrices, and qualification checklists are retained for a minimum of 5 years, stored securely, and accessible for audits.

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.

9.2. Revised documents require review and approval before implementation.

9.3. Old versions must be archived and marked as "Obsolete."

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Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
B				

26.0

1. Purpose

1.1. Ensure that manufacturing and processes are planned, executed, monitored, and controlled to consistently produce conforming product.

2. Scope

2.1. Applies to all production, assembly, and process-related activities.

3. References

3.1. ISO 9001
3.2. GMP Guidelines
3.3. Company Quality Management System (QMS)

4. Definitions

4.1. Process Parameter: A measurable factor that must be controlled to ensure process consistency (e.g., temperature, pressure, speed).
4.2. Process Validation: Documented evidence that a process consistently yields acceptable product.
4.3. Critical Process: Any step where a deviation could result in nonconforming product or safety risk.

5. Responsibilities

5.1. Production Manager: Authorize and maintain all SOPs/work instructions, ensure operators are trained and processes are validated.
5.2. Process Owners: Define critical parameters and acceptance criteria, monitor process performance and initiate corrective action when out of tolerance.
5.3. Operators: Follow documented instructions, record process data, report abnormalities immediately.
5.4. Quality assurance: Review process records, trend data, and approve and process changes.

6. Procedure

6.1. Process Documentation
  6.1.1. Develop and approve SOPs and work instructions for each process step, including tooling setups, machine parameters, and safety precautions.
6.2. Process Validation (for critical processes)
  6.2.1. Create Validation Protocol: define acceptance criteria, sample size, and test methods.
  6.2.2. Perform Validation Runs under normal operating conditions.
  6.2.3. Document results in an Inspection Report; obtain QA approval before routine production.
6.3. Setup and Run
  6.3.1. Verify equipment is within calibration and approved maintenance status.
  6.3.2. Perform a first-piece inspection and document on the First-Article Inspection Report.
6.4. In-Process Monitoring
  6.4.1. At predefined intervals or sample sizes, measure key parameters/characteristics and record on the In-Process Inspection Checklist.

6.4.2. Compare measured values to acceptance criteria; hold production if out of tolerance.
6.4.3. Investigate and resolve any deviations via NCR/CAPA.

6.5. Control of Process Changes
6.5.1. Any adjustment to process parameters must follow the Document Control procedure (P-01).
6.5.2. Evaluate the impact of changes through risk assessment and, if necessary, re-validate the process.

7. Supporting Documents

7.1. SOP / Work Instruction template.
7.2. Process Flow Diagram.
7.3. Validation Protocol & Report Templates.
7.4. Process Setup Log.
7.5. First-Article Inspection Report.
7.6. In-Process Inspection Checklist.

8. Recordkeeping

8.1. Production Batch Records: Retain for a minimum of 3 years in.
8.2. Validation Reports: Retain for life of the product + 2 years in “04_Records/Validation.”
8.3. In-Process Inspection Checklists & Setup Logs: Retain for 1 year in.
8.4. All records must be legible, signed, and dated; inaccessible to unauthorized changes.

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.
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1. **Purpose**
   1.1. Define how customer inquiries, quotes, orders, and changes are reviewed, accepted, processed, and communicated to ensure requirements are met.
2. **Scope**
   2.1. From customer inquiry through shipment and invoicing for products and services.
   2.2. Includes contract review, order entry, changes, cancellations, and special requirements.
3. **References**
   3.1. ISO 9001:2015 Section 8.2 (Requirements for products and services)
4. **Definitions**
   4.1. Contract Review: Verification that requirements are defined, understood, and can be met before acceptance.
   - Special Requirements: Critical characteristics, regulatory, safety, or customer unique clauses.
5. **Responsibilities**
   5.1. Sales: Manage inquiries/quotes; perform initial contract review; confirm acknowledgments.
   5.2. Order Entry: Create/maintain Sales Orders; ensure accurate data and revisions.
   5.3. Finance: Credit checks, tax status, billing terms, credit holds.
   5.4. Planning: Convert Sales Orders to work orders, routing, and material requirements.
   5.5. Quality: Review special/regulated requirements and key characteristics.
6. **Procedure**
   6.1. Inquiry & Quote: Capture requirements (drawings/revisions, specs, quantities, dates, delivery terms, quality clauses); validate feasibility; issue formal quote with validity period.
   6.2. Order Receipt & Contract Review: Compare PO to quote (price, qty, revision, terms, delivery, quality requirements); resolve discrepancies; finance performs credit check.
   6.3. Order Entry & Acknowledgment: Create Sales Order; attach controlled drawings/specs and customer PO; assign promise date; send acknowledgment.
   6.4. Release to Operations: Planning creates work orders, routing, travelers, and picks; notify Purchasing; Quality flags special characteristics and required inspections.
   6.5. Order Changes/Cancellations: Log change request; re-review feasibility/impact; update SO and travelers; issue revised acknowledgment; reconcile costs on cancellations.
   6.6. Records & Communication: Store all controlled job documents; communicate delays/risks proactively with revised promise dates.

1.

7. Supporting Documents

7.1. Quote Worksheet
7.2. Order Review Checklist
7.3. Sales Order Acknowledgment
7.4. Order Change Log

8. Recordkeeping

8.1. All POs, SOs, acknowledgments, change logs, and controlled attachments are retained for a minimum of 5 years and organized by job folder for traceability.

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.
9.2. Revised documents require review and approval before implementation.
9.3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release	J Horanburg		9/10/25
A				
B				

1. Purpose

1.1. The purpose of this Procedure is to establish a structured approach for the development of new products. This is a high level and rapid review of the business case, competitive landscape, product requirements and production capabilities to determine value to the company. The result shall be a decision to proceed, postpone or cease new product development activities. A decision to proceed will advance new product development to the Design Control Procedure within the Quality Management System (QMS). This ensures product safety, performance, and compliance with regulatory requirements.

2. Scope

2.1. This procedure applies to new product development. A flow chart of the procedure is illustrated in Appendix A.

3. References

3.1. ISO 9001:2015 Section 8.3 (Design and Development)
3.2. GMP Guidelines
3.3. Company Quality Management System (QMS)
3.4. Design Control

4. Definitions

5. Responsibilities

5.1. Management: Responsible for the overall conformance to the company business plan.
5.2. Sales: Responsible for defining a business case.
5.3. Engineering: Responsible for defining product and process specifications which, when applied, shall achieve the business case.
5.4. All above are responsible for deciding to proceed, postpone or cease future development activity.

6. Procedure

6.1. Management determines the need for a new product.
6.2. Develop the business case.
6.2.1. Size of market/opportunity - $$ and pieces.
6.2.2. Who are the customers?
6.2.3. Who is the competition?
6.2.4. How do we compete?
6.2.5. What are the product attributes and specifications that allow us to compete (ie. What does the product look like).
6.2.6. Sale Price points.
6.2.7. How do we enter the market?
6.2.8. Reasonable 1–3 year annual sales projection.

6.3. Develop the Preliminary Design (product and process specifications to support the business case).
  6.3.1. Raw Materials.
  6.3.2. Dimensions.
  6.3.3. Mechanical and chemical characteristics.
  6.3.4. Manufacturing process to be used.
  6.3.5. Development capital and expense costs.
6.4. Financial analysis.
  6.4.1. Considering sales and cost estimates, determine Return on Investment (ROI).
6.5. Conduct SWOT analysis.
  6.5.1. Identify business risks.
  6.5.2. Identify risk mitigation plan.
  6.5.3. Decide if new product development activity shall proceed, postpone or cease.
  6.5.4. If Proceeding, all future activities shall conform to the company Design Control Procedure.
  6.5.5. Initial GO/NO-GO Decision.

7. Procedure (Phase 2) – Product Design

7.1. Sales/Marketing Team tasks.
  7.1.1. Customer input.
  7.1.2. Marketing Plan.
  7.1.3. Spec sheets and installation instructions (coordinate w/ Engineering).
  7.1.4. Sales plan.
7.2. Engineering Team Tasks – Ensure adherence to Design and Document Control.
  7.2.1. Model & Design Product and Process.
  7.2.2. Model & design tooling.
  7.2.3. Project Schedule and Budget.
  7.2.4. Design, build, test final product.

8. Procedure (Phase 3) – Release Product for Production

8.1. Product print, spec sheets and installation instructions.
8.2. Process settings and procedures.
8.3. Operator Training if required.
8.4. Ensure all product and process documents are released under the requirements of Document Control.

9. Supporting Documents

9.1. Business case summary.
9.2. Preliminary product specification.
9.3. SWOT analysis.
9.4. Risk mitigation summary.
9.5. Financial analysis with ROI.

10. Recordkeeping

10.1. Records will be stored electronically within the Quality Management System for a minimum of three (3) years.

10.2. Obsolete documents will be archived for historical reference.

11. Revision History

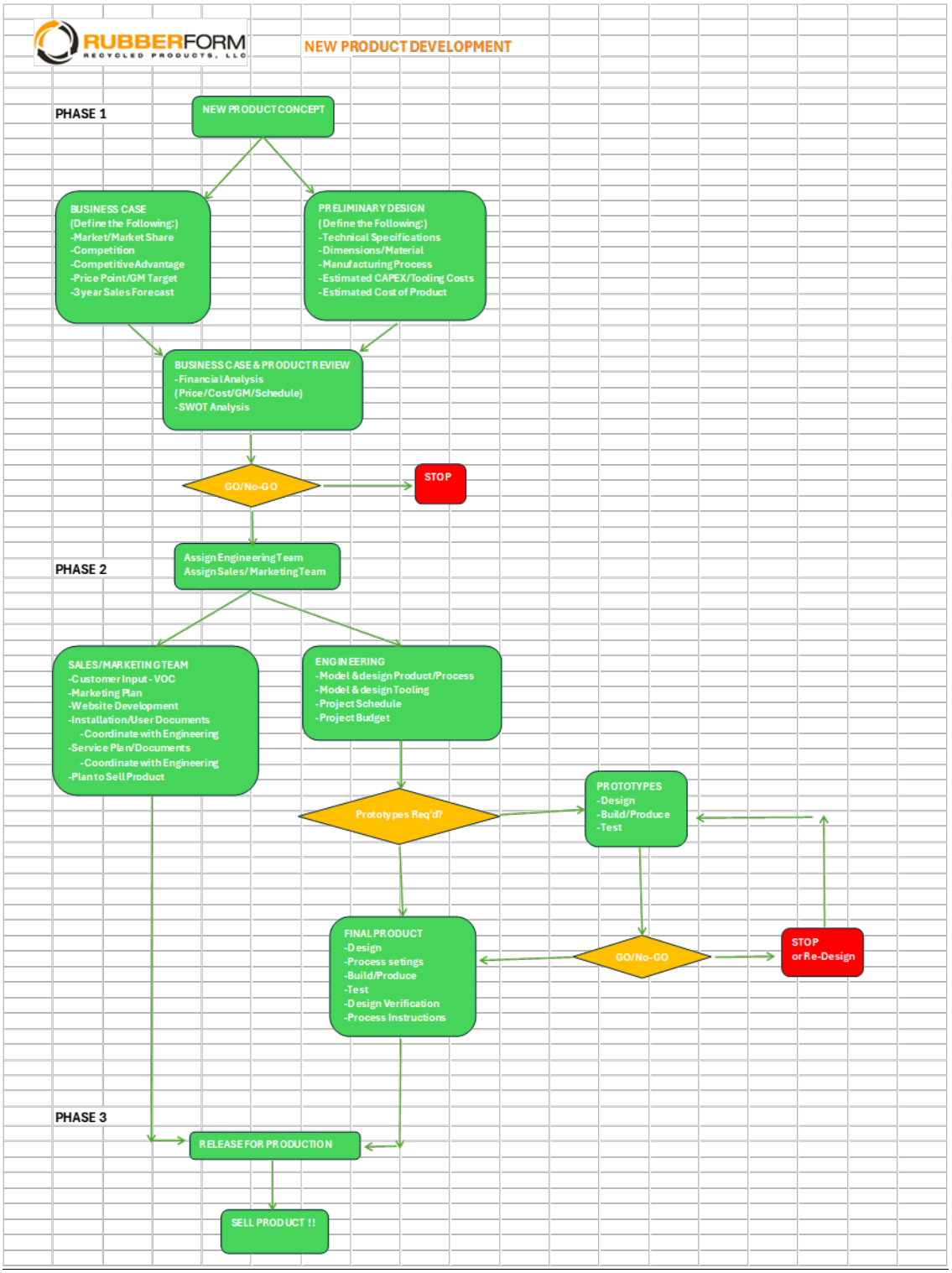
11.1. Any modifications must be documents with a revision history, including changes made and the reason for the update.

11.2. Revised documents require review and approval before implementation.

11.3. Old versions must be archived and marked as “Obsolete”.

APPENDIX – A (See next page)

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
B				


RUBBERFORM RECYCLED PRODUCTS, LLC
NEW PRODUCT DEVELOPMENT
PHASE 1
NEW PRODUCT CONCEPT
BUSINESS CASE
(Define the Following:)
-Market/Market Share
-Competition
-Competitive Advantage
-Price Point/GM Target
-3 year Sales Forecast
PRELIMINARY DESIGN
(Define the Following:)
-Technical Specifications
-Dimensions/Material
-Manufacturing Process
-Estimated CAPEX/Tooling Costs
-Estimated Cost of Product
BUSINESS CASE & PRODUCT REVIEW
-Financial Analysis
(Price/Cost/GM/Schedule)
-SWOT Analysis
GO/No-GO
STOP
PHASE 2
Assign Engineering Team
Assign Sales/Marketing Team
SALES/MARKETING TEAM
-Customer Input - VOC
-Marketing Plan
-Website Development
-Installation/User Documents
-Coordinate with Engineering
-Service Plan/Documents
-Coordinate with Engineering
-Plan to Sell Product
ENGINEERING
-Model & design Product/Process
-Model & design Tooling
-Project Schedule
-Project Budget
Prototypes Req'd?
PROTOTYPES
-Design
-Build/Produce
-Test
FINAL PRODUCT
-Design
-Process setings
-Build/Produce
-Test
-Design Verification
-Process Instructions
GO/No-GO
STOP
or Re-Design
PHASE 3
RELEASE FOR PRODUCTION
SELL PRODUCT !!

1. Purpose

1.1. Ensure purchased products and services meet specified requirements through supplier selection, purchasing controls, verification, and performance monitoring.

2. Scope

2.1. All externally provided products/services including raw materials, components, subcontracting, calibration, and logistics.

3. References

3.1. ISO 9001:2015 Section 8.4 (Control of externally provided processes, products, and services).
3.2. Receiving Inspection (P-10).
3.3. Nonconforming Outputs (P-11).
3.4. Inventory Control (P-08.1)

4. Definitions

4.1. Critical Supplier: Directly affects product conformity/safety or regulatory compliance.
4.2. SCAR: Supplier Corrective Action Request.

5. Responsibilities

5.1. Purchasing: Qualify suppliers; issue POs; flow down requirements; monitor performance.
5.2. Quality: Approve critical suppliers; define verification criteria; manage SCARs.
5.3. Receiving: Verify incoming items per PO/specs and record results.
5.4. Requestors/Engineers: Define specifications, drawings, and special characteristics.

6. Procedure

6.1. Supplier Selection & Approval: Evaluate new suppliers (capability, certifications, capacity, OTD history, pricing, risks); Quality approves critical suppliers; add to ASL.
6.2. Purchasing Controls: Issue PO with complete requirements including rev level, quality clauses, certificates, packaging, and delivery terms; changes require revised PO and supplier acknowledgment.
6.3. Verification of Purchased Product: Receiving inspects identity, count, damage, certs, and critical characteristics/tests when specified; process nonconforming material per P 11; maintain evidence for dock to stock or delegated inspection.
6.4. Supplier Performance Monitoring: Track OTD, quality (PPM/defects), responsiveness; review at least quarterly; issue SCARs for systemic issues; remove from ASL if unresolved.

7. Supporting Documents

7.1. Supplier Evaluation.
7.2. Emergency Buy Risk Assessment.
7.3. Supplier Scorecard (F-09-03)
7.4. Purchase Orders, Certifications, Inspection Records.

8. Recordkeeping

8.1. Supplier evaluations
8.2. Inspection Report
8.3. PO and Drawing sent to vendor

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.
9.2. Revised documents require review and approval before implementation.
9.3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
B				

1. Purpose

1.1. Ensure that all incoming materials, in-process components, and finished products conform to specified requirements before release.

2. Scope

2.1. Applies to:

2.1.1. Incoming inspections of purchased materials/parts.

2.1.2. In-process inspections during manufacturing.

2.1.3. Final product inspections prior to shipment.

2.2. Excludes:

2.2.1. Equipment calibration activities (see Calibration & Maintenance).

2.2.2. Destructive testing not routinely performed.

3. References

3.1. ISO 9001

3.2. GMP Guidelines

3.3. Company Quality Management System (QMS)

4. Definitions

4.1. Acceptance Criteria: The documented tolerances or measurements that a product or part must meet.

4.2. Hold Area: A designated location for product pending inspection or disposition.

4.3. Lot: A defined quantity of product manufactured under uniform conditions.

4.4. WIP: Work in progress

5. Responsibilities

5.1. QC Inspector: Performs and documents all inspections/tests.

5.2. Production Supervisor: Ensures items are presented for inspection and enforces hold-area compliance.

5.3. Quality Manager: Reviews inspection data trends and authorizes final release.

6. Procedure

6.1. Preparation

6.1.1. Verify that inspection equipment is calibrated and within its calibration period.

6.1.2. Obtain the latest drawing/specification and Inspection Checklist for the item or lot.

6.2. Incoming Inspection (Vendor supplied materials)

6.2.1. Record supplier name, part number, and lot/date code.

6.2.2. Inspect a sample per the Sampling Plan (e.g., ANSI/ISO AQL).

6.2.3. Compare measurements to Acceptance Criteria.

6.2.4. Document results on the Incoming Inspection Report.

6.2.5. Move accepted lots to “Released Materials” area; tag any nonconforming lots and notify QA.

6.3. In-Process Inspection (RubberForm Work-In-Progress)

6.3.1. At predefined process stages (e.g., after machining, assembly), inspect per In-Process Checklist.

6.3.2. Record measurements and observations.

6.3.3. If nonconformance is detected, stop the line, tag parts, and initiate NCR/CAPA.

6.4. Final Product Inspection (RubberForm Product)

6.4.1. Sample items per Final Inspection Plan.

6.4.2. Perform all dimensional, functional, and cosmetic checks.

6.4.3. Complete the Final Inspection Report, noting any deviations.

6.4.4. Release product for shipment by signing the “Release for Delivery” section; hold any rejects for disposition.

6.5. Disposition of Nonconforming Materials/WIP/Products

6.5.1. Segregate nonconforming items in the Hold Area.

6.5.2. Complete a Nonconformance Report (NCR) and route for disposition (rework, scrap, return to supplier).

7. Supporting Documents

7.1. Inspection Checklist (Incoming, In-Process, Final)

7.2. Sampling Plan & Acceptance Criteria Matrix

7.3. Inspection/Test Protocols (detailed test methods)

7.4. Incoming & Final Inspection Report Forms

7.5. Nonconformance Report (NCR) Form

8. Recordkeeping

8.1. Inspection Reports: retained for 3 years (or as required by customer/regulation).

8.2. Calibration Certificates: filed with the equipment records for the life of the asset.

8.3. NCR Records: retained per CAPA retention rules (minimum 5 years).

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.

9.2. Revised documents require review and approval before implementation.

9.3. Old versions must be archived and marked as "Obsolete."

26.0

Notes:

- All records shall be legible, dated, and signed.
- Review inspection trends quarterly to drive continuous improvement.

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
B				

26.0

1. Purpose

1.1. Define a consistent method to capture, triage, investigate, resolve, and prevent recurrence of customer complaints.

2. Scope

2.1. All customer reported issues including quality, delivery, documentation, and service.

3. References

3.1. ISO 9001:2015 Section 9.1.2 (Customer satisfaction) & 8.7 (Control of nonconforming outputs)

3.2. Nonconforming Outputs , Corrective Action (CAPA) , RMA Work Instruction (WI RMA)

4. Definitions

4.1. Complaint: Any expression of dissatisfaction related to company products or services.

4.2. Severity Levels:

4.2.1. S1: Safety/regulatory.

4.2.2. S2: Functional performance.

4.2.3. S3: Cosmetic/documentation.

4.2.4. S4: Inquiry

5. Responsibilities

5.1. Customer Service: Log complaints; acknowledge receipt; coordinate communication.

5.2. Quality: Triage severity; lead investigation; define containment/corrective actions.

5.3. Department Managers: Implement containment and corrective actions; provide resources.

5.4. Sales: Manage customer expectations, commercial terms, replacements/credits.

6. Procedure

6.1. Intake & Acknowledgment: Log complaint with PO/lot/serial, description, photos, dates; acknowledge within 1 business day with a NCR and next steps.

6.2. Triage & Containment: Assign severity S1–S4; for S1 initiate immediate containment and management notification; options include stop ship, stock purge, field alert, replacement, rework, or credit hold.

6.3. Investigation: Collect evidence (production records, inspections, supplier certs, training, maintenance, change logs); perform root cause analysis, document results.

6.4. Corrective & Preventive Action: If systemic, open CAPA; define actions, owners, due dates, and verification method; verify effectiveness via audit/capability/trend data.

6.5. Customer Response & Closure: Provide updates; issue formal response with findings, cause, actions, and prevention; close upon effectiveness verification and customer acceptance where required.

6.6. Returns (RMA): Issue RMA number and instructions; inspect returned items; update investigation and disposition.

7. Supporting Documents

7.1. Complaint Log
7.2. NCR Form
7.3. RMA records, test/inspection reports, communications

8. Recordkeeping

8.1. Complaint logs, investigations, responses, RMAs, and CAPA records are retained for a minimum of 5 years; trends are reviewed periodically for continuous improvement.

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.
9.2. Revised documents require review and approval before implementation.
9.3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
B				

1. Purpose

1.1. The purpose of this Non-Conformances Procedure is to establish a structured approach for identifying, documenting, investigating, and resolving non-conforming products manufactured by Rubberform. This ensures that defective or non-conforming products are properly controlled to prevent unintended use or delivery, and that appropriate corrective actions are taken to prevent recurrence and improve product quality.

2. Scope

2.1. This procedure applies to all non-conforming products identified in manufacturing, inspections, customer feedback, and supplier-provided materials, including but not limited to:

2.1.1. Manufacturing defects and process deviations.

2.1.2. Supplier-provided material non-conformances.

2.1.3. Customer complaints related to product quality.

2.1.4. Internal audit findings and regulatory compliance gaps.

2.2. This procedure does not apply to minor issues that do not impact product quality, compliance, or safety and can be resolved immediately.

3. References

3.1. ISO 9001

3.2. GMP Guidelines

3.3. Company Quality Management System (QMS)

4. Definitions

4.1. Non-Conformance (NC): Any deviation from approved specifications, standards, or procedures that effects product quality, safety, or compliance.

4.2. Correction: An immediate action taken to address a detected non-conformance.

4.3. Corrective Action: A long-term action taken to eliminate the root cause of a non-conformance and prevent recurrence.

4.4. Containment: Temporary measures implemented to isolate and control non-conforming products or processes.

5. Responsibilities

5.1. Employees: Identify and report non-conformances promptly.

5.2. Quality Assurance Team: Investigate non-conformances, determine root causes, and oversee corrective actions.

5.3. Production & Operations Team: Implement containment actions and corrective measures.

5.4. Management: Approve corrective actions and provide resources for resolution.

6. Procedure

6.1. Identification & Reporting

26.0

6.1.1. Identify non-conforming products through inspections, audits, customer feedback, or process monitoring.
6.1.2. Document the non-conformance using the Non-Conformance Report (NCR) Form.
6.1.3. Notify the Quality Assurance Team for further investigation.

6.2. Containment & Immediate Actions
6.2.1. Segregate and label non-conforming products or materials to prevent unintended use.
6.2.2. If applicable, suspend affected processes to avoid further deviations.
6.2.3. Apply temporary corrective measures to minimize impact.

6.3. Investigation & Root Cause Analysis
6.3.1. Conduct a Root Cause Analysis (RCA).
6.3.2. Gather data from process records, inspections, and personnel interviews.
6.3.3. Determine the underlying cause of the non-conformance.

6.4. Corrective & Preventive Actions
6.4.1. Develop a CAPA plan outlining actions to eliminate root causes.
6.4.2. Assign responsibility and deadlines for implementation.
6.4.3. Implement preventive measures to avoid recurrence, such as process changes, retraining, or supplier evaluations.

6.5. Verification & Effectiveness Check
6.5.1. Monitor corrective actions to ensure proper implementation.
6.5.2. Conduct follow-up inspections or audits to verify effectiveness.
6.5.3. Update relevant procedures, training materials, or documentation as needed.

6.6. Closure & Recordkeeping
6.6.1. Document all actions taken, including investigation findings and resolutions, in the Non-Conformance Log.
6.6.2. Retain records as per company retention policies.
6.6.3. Review trends in non-conformances periodically to identify areas for continuous improvement.

7. Supporting Documents

7.1. Appendix A: Non-Conformance Report (NCR) Form
7.2. Appendix B: Corrective Action Plan Template

8. Recordkeeping

8.1. All non-conformance records, investigation reports, and corrective action plans must be retained for a minimum of 5 years.
8.2. The Non-Conformance Log must be reviewed periodically to identify recurring issues and trends.
8.3. Records must be securely stored and accessible for audits and regulatory inspections.

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.

9.2. Revised documents require review and approval before implementation.
9.3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
B				

1. Purpose

1.1. The purpose of this Corrective Action and Preventive Action (CAPA) Procedure is to establish a structured approach for identifying, documenting, investigating, and resolving quality issues within the Quality Management System (QMS). This ensures continuous improvement by addressing root causes and preventing recurrence of issues.

2. Scope

2.1. This procedure applies to all corrective and preventive actions initiated due to:

- 2.1.1. Non-conformances identified in manufacturing, inspections, or audits.
- 2.1.2. Customer complaints related to product quality.
- 2.1.3. Supplier-related defects.
- 2.1.4. Process deviations and equipment failures.
- 2.1.5. Regulatory compliance issues.

2.2. This procedure does not apply to minor issues that can be immediately corrected without further investigation.

3. References

3.1. ISO 9001
3.2. GMP Guidelines
3.3. Company Quality Management System (QMS)

4. Definitions

4.1. Corrective Action (CA): A long-term action taken to eliminate the root cause of an identified issue and prevent recurrence.
4.2. Preventive Action (PA): A proactive measure taken to prevent potential non-conformances before they occur.
4.3. Root Cause Analysis (RCA): A systematic method used to determine the underlying cause of a problem.
4.4. CAPA Log: A centralized record of all corrective and preventive actions taken within the organization.

5. Responsibilities

5.1. Employees: Identify and report potential issues requiring CAPA.
5.2. Quality Assurance Team: Investigate issues, determine root causes, and oversee CAPA implementation.
5.3. Production & Operations Team: Implement corrective and preventive actions as assigned.
5.4. Management: Approve CAPA plans and provide necessary resources for resolution.

26.0

6. Procedure

6.1. Issue Identification & Reporting

6.1.1. Identify quality issues through non-conformance reports, audits, customer complaints, or process monitoring.

6.1.2. Document the issue using the CAPA Request Form.

6.1.3. Submit the request to the Quality Assurance Team for review.

6.2. Investigation & Root Cause Analysis

6.2.1. Assign a CAPA owner responsible for investigation.

6.2.2. Conduct a Root Cause Analysis (RCA).

6.2.3. Gather relevant data, including process records and personnel feedback.

6.2.4. Determine whether a corrective or preventive action is required.

6.3. Corrective Action Implementation

6.3.1. Develop a Corrective Action Plan (CAP) outlining steps to eliminate the root cause.

6.3.2. Assign responsibilities and deadlines for action items.

6.3.3. Implement process changes, retraining, or supplier corrective actions as needed.

6.3.4. Monitor the implementation progress.

6.4. Preventive Action Implementation

6.4.1. Identify potential risks based on trend analysis of past issues.

6.4.2. Implement proactive process controls or equipment upgrades.

6.4.3. Train employees on preventive measures.

6.5. Verification & Effectiveness Check

6.5.1. Conduct follow-up audits or inspections to verify CAPA effectiveness.

6.5.2. Assess if corrective and preventive actions successfully eliminated or reduced the risk.

6.5.3. Update procedures, training materials, or records as necessary.

6.6. Closure & Recordkeeping

6.6.1. Document all actions taken, including investigation findings and resolutions, in the CAPA Log.

6.6.2. Retain records per company retention policies.

6.6.3. Review CAPA trends periodically to identify opportunities for continuous improvement

7. Supporting Documents

7.1. Appendix A: CAPA Request Form

7.2. Appendix B: Root Cause Analysis Worksheet

7.3. Appendix C: Corrective Action Plan Template

7.4. Appendix D: Preventive Action Plan Template

8. Recordkeeping

8.1. All CAPA records, investigation reports, and action plans must be retained for a minimum of 5 years.

8.2. The CAPA Log must be reviewed periodically to track trends and assess effectiveness.

8.3. Records must be securely stored and accessible for audits and regulatory inspections.

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.

9.2. Revised documents require review and approval before implementation.

9.3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
B				

1. Purpose

1.1. To define a standardized process for initiating, executing, documenting, and closing work orders, to ensure traceability, equipment reliability, and ISO 9001 compliance.

2. Scope

2.1. This procedure applies to all maintenance activities at RubberForm including:
- 2.1.1. Preventive Maintenance (PM)
- 2.1.2. Corrective Maintenance (CM)
- 2.1.3. Predictive Maintenance (PdM)
- 2.1.4. Emergency Maintenance (EM)

2.2. This procedure covers both internal maintenance tasks and those assigned to external service.

3. References

3.1. ISO 9001
3.2. GMP Guidelines
3.3. Company Quality Management System (QMS)
3.4. CMMS (Computerized Maintenance Management System)
3.5. Facility Safety Procedures

4. Definitions

4.1. Work Order (WO): A unique, traceable document or digital entry that authorizes maintenance work to be performed.
4.2. CMMS: Computerized system used to schedule, track, and log maintenance activities.
4.3. PM: Preventive Maintenance: scheduled maintenance to reduce failures.
4.4. CM: Corrective Maintenance: repair work due to equipment failure or malfunction.
4.5. PdM: Predictive Maintenance — condition-based monitoring to predict and prevent failures.
4.6. EM: Emergency Maintenance — immediate repair due to safety or production stoppage.

5. Responsibilities

5.1. Requestor (Operator, Supervisor, or Quality Inspector)
- 5.1.1. Initiates the work order request through CMMS or paper form.
- 5.1.2. Provides detailed information regarding the issue or request.

5.2. Production Manager
- 5.2.1. Reviews and prioritizes incoming work orders.
- 5.2.2. Assigns work based on urgency, technician availability, and production schedule.
- 5.2.3. Ensures safety precautions and Lock-Out Tag-Out (LOTO) procedures are followed.

5.3. Maintenance Technician
- 5.3.1. Reviews assigned work orders before starting.
- 5.3.2. Gathers required tools, parts, and PPE.
- 5.3.3. Executes the task safely and completely.
- 5.3.4. Completes the work order with notes, labor time, parts used, and resolution code.

5.4. Maintenance Manager

5.4.1. Verifies work order completion and closes the ticket in CMMS.
5.4.2. Reviews data for trends, recurring issues, and areas for improvement.
5.4.3. Ensures proper documentation is retained for audit readiness.

6. Procedure

6.1. Work Order Initiation
6.1.1. Work order is initiated via digital maintenance request form or paper form when an issue is observed or a scheduled PM is due.
6.1.2. Requestor includes machine ID, location, issue description, urgency, and photos if applicable.
6.2. Work Order Review and Scheduling
6.2.1. Maintenance Lead reviews all open work orders daily.
6.2.2. Classifies priority:
6.2.2.1. P1: Emergency (stop production/safety hazard)
6.2.2.2. P2: Urgent (affects production speed or quality)
6.2.2.3. P3: Routine
6.2.3. Schedules task based on availability and impact.
6.3. Task Execution
6.3.1. Technician retrieves assigned work orders at start of shift or via real-time system.
6.3.2. Performs task per documented SOP or troubleshooting guide.
6.3.3. Follows all safety and lockout/tagout procedures.
6.3.4. Notes any corrective actions, part replacements, or findings.
6.4. Work Order Completion and Documentation
6.4.1. Technician logs labor hours, materials used, and resolution in the work order.
6.4.2. If task is not completed, technician indicates status and reason (e.g., awaiting part).
6.4.3. Maintenance Lead reviews and closes the work order once verified.
6.5. Record Retention and Reporting
6.5.1. Closed work orders are retained per document control requirements (typically 3 years).
6.5.2. Monthly reports summarize downtime causes, PM compliance, and technician efficiency.
6.5.3. Trends are reviewed as part of management review and continual improvement.

7. Supporting Documents

7.1. N/A

8. Recordkeeping

8.1. To be kept:
8.1.1. Maintenance Work Orders (digital or paper).
8.1.2. Technician Logs.
8.1.3. Downtime Reports.
8.1.4. Preventive Maintenance Schedules.
8.1.5. Maintenance Performance KPIs.

9. Revision History

9.1. Any modifications must be documented with a revision history, including changes made and the reason for the update.

9.2. Revised documents require review and approval before implementation.

9.3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
B				

Directions: This form is used to document relevant information related to the RubberForm LLC Quality Management System (QMS). Accuracy of this information may affect design control, product quality, customer satisfaction and, ultimately, financial performance of the company. Please provide as much information as possible and complete it as accurately as possible.

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release	J Horanburg	J Probst	10/1/2025
A				
B				

Instructions:

1. Fill out this form to create a new QMS controlled document or change an existing QMS document.
2. Describe justification for the requested change.
3. Secure approval of two or more management positions authorized to approve document changes.
4. Submit completed form appropriate person to make the change.

Person Requesting the Change:
Date:

Document Title:

Document Number & Current Revision:

Justification or Reason for Change:

Approval Signatures (at least two) and Date:
Management:
Production:
Engineering:
Quality:
Sales:

Person Completing the Change:
Date:

26.0

Directions: This form is used to document relevant information related to the RubberForm LLC Quality Management System (QMS). Accuracy of this information may affect design control, product quality, customer satisfaction and, ultimately, financial performance of the company. Please provide as much information as possible and complete it as accurately as possible. Attach additional information as necessary.

PHASE 1: NEW PRODUCT CONCEPT

Business Case:

Define Target Market:
Click or tap here to enter text.
Who is the competition:
Click or tap here to enter text.
Define RubberForm competitive advantage:
Click or tap here to enter text.
Price point/GM Target:
Click or tap here to enter text.
3 year sales forecast:
Click or tap here to enter text.

Preliminary Design:

Technical Specifications:
Click or tap here to enter text.
Dimensions/Materials:
Click or tap here to enter text.
Manufacturing Process:
Click or tap here to enter text.
Estimated CAPEX”
Click or tap here to enter text.
Estimated Cost of Product
Click or tap here to enter text.

Refer to New Product Development Procedure for disposition of Phase 1

PHASE 2: DETAILED NEW PRODUCT DESIGN & DEVELOPMENT PLAN

(Refer to New Product Development Procedure and Design Control Procedure)

Sales & Marketing Actions:

Action Description	Who
Customer input requirements:	
Click or tap here to enter text.	
Marketing Plan:	
Click or tap here to enter text.	
Install/User Documents:	
Click or tap here to enter text.	
Service Plan Documents:	
Click or tap here to enter text.	
Sales Implementation Plan:	
Click or tap here to enter text.	

Engineering Actions:

Action Description	Who
Product & Process Design:	
Click or tap here to enter text.	
CAD Models (incl FEA):	
Flexi-Base Assy.sldasm Sign Base Assy.sldasm Bollard Cover.sldasm	
Die & Tooling Design:	
Click or tap here to enter text.	
Process Development Planning:	
Click or tap here to enter text.	
Project Budget & Schedule:	
Click or tap here to enter text.	
Project Management if Required:	
Click or tap here to enter text.	
Process settings & work instructions:	
Click or tap here to enter text.	
Design Verification and Validation:	
Click or tap here to enter text.	
Training Requirements:	
Click or tap here to enter text.	

26.0

PHASE 3: PRODUCTION RELEASE

(Refer to New Product Development Procedure and Design Control Procedure)

Production Release Checklist:

COMPLETED ITEM	WHO	APPROVED	NOT APPROVED	N/A
Manufacturing Drawings				
Process Settings				
Work Instructions				
Operator Training				
Maintenance Training				
In Process Documents & Records				
Production Control Documents (SKU, Inspection, Inventory, etc)				
Customer Product Print/Specs				
Hardware, Sealants, Other Items Provided to Customer				
Customer/User Installation Directions				
Sales & Marketing Plan				
Production Samples				
Work cell				
Process Equipment				
Packaging				

Form Revision History

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release	J Horanburg	J Probst	10/1/2025
A				
B				

Who wants it	
What is wanted	
Where is it going to be used	
When do you need it	
Why do you not have it yet	
How does it help your day	

F-10	RUBBERFORM RECYCLED PRODUCTS, LLC	What:	Incoming Inspection
		Where:	Receiving
		Who:	All
	File Location		

Who Inspected	
What was Inspected	
Where is it now	
When did we receive it	
Why was it ordered	
How do we know it is good?	

Submit Form to: Quality Administrator		Created:	6/22/25
		Released:	10/01/2025
		Revised:	[revision date]
		Approved:	[owners signature]

26.0

Aesthetic

Spec. Location	Spec. Requirement	Actual Measured	Deviation from Spec?

Dimensional

Drawing #			
Dim Tolerance	**Required Dim.**	**Actual Dim.**	**Deviation from Required**

Function

Literature #			
Set Expectation	**Actual Performance**		

Additional Inspection Points

Ref. Doc.			
What	**Where**	**How**	**Why**

Who	
What	
Where	
When	
Why	
How	

F-13	RUBBERFORM RECYCLED PRODUCTS, LLC	What:	NCR Form
		Where:	All Departments
		Who:	All Employee's
	File Location	C:\Users\jesse\Documents\RUBBERFORM\QMS\CURRENT MASTER QMS\MASTER\Forms\F-13 Non-Conformance Form.docx	

NCR # 25_ _ _

Vendor		**Part # / Process**	
P.O. #		Description	
Qty Effected		Drawing #	
Packing list #		Rev #	

Description of Nonconformance:

Requirement:	**Actual:**

Identified by..... **Date: / / 2025**

Disposition:

RETURN TO VENDOR	(QTY):	VENDOR REWORK	(QTY):
REQUEST CREDIT	**(QTY):**	IN-HOUSE REWORK	(QTY):
USE AS IS -	(QTY):	SCRAP	(QTY):

Immediate Action Taken:

Determined by: . . . . . Date: / / 2025

IS CAPA INVESTIGATION NECESSARY?

Circle One YES / NO If NO please explain

Comments:

Approved by: . . . . . Date: / / 2025

REVIEW AND APPROVAL OF NONCONFORMANCE DISPOSITION AND CAPA DETERMINATION

Comments:

Approved by: . . . . . Date: / / 2025

Submit Form to: Production Manager		Created:	6/22/25
		Released:	8/28/25
		Revised:	[revision date]
		Approved:	[owners signature]

F-13	RUBBERFORM RECYCLED PRODUCTS, LLC	What:	NCR Form
		Where:	All Departments
		Who:	All Employee's
	File Location	Z:\QMS CURRENT COPY\MASTER\FORMSC:\Users\jesse\Documents\RUBBERFORM\QMS\CURRENT MASTER QMS\MASTER\Forms\F-13e Non-Conformance Form.docx	

Date Initiated?	Click or tap to enter a date.
Who Initiated?	Click or tap here to enter text.
What is the Problem?	Click or tap here to enter text.
What Department?	
Where does it effect?	Choose an item.
When did it happen?	Click or tap to enter a date.
How many times?	Click or tap here to enter text.
Mission Critical?	Yes ☐ No ☐

What Happened? Requirement – Actual – Pictures

What is Required?	What is Actual?
Click or tap here to enter text.	Click or tap here to enter text.

Assembly, Product, Part, Component Details

Vendor	Click or tap here to enter text.
Purchase Order (PO) #	Click or tap here to enter text.
Quantity Effected	Click or tap here to enter text.
Notes	Click or tap here to enter text.
Product Location	Click or tap here to enter text.
Rejection Tag #	Click or tap here to enter text.
Submitted to	Click or tap here to enter text.
Disposition	Click or tap here to enter text.

IS CAPA INVESTIGATION NECESSARY?

Circle One **YES / NO**

Submit Form to: Supervisor or Department Manager		Created:	6/22/25
		Released:	6/26/25
		Revised:	[revision date]
		Approved:	[owners signature]

26.0

Who	
What	
Where	
When	
Why	
How	

26.0

F-14

RUBBERFORM
RECYCLED PRODUCTS, LLC

What:	Equipment Status
Where:	
Who:	

Purpose:	Provide data collection point for Managers and Operators

Machine Name	
Machine ID	
Department	
Yrs in Service	
Skill Level	

PARTS NEEDED;

| |_____	$_____
| |_____	$_____
| |_____	$_____
| |_____	$_____
| |_____	$_____
| |_____	$_____
| |_____	$_____
| |_____	$_____
| |_____	$_____

SPECIALIST NEEDED:

| |_____	Hrs_____
| |_____	Hrs_____
| |_____	Hrs____
| |_____	Hrs____
| |_____	Hrs____
| |_____	Hrs____
| |_____	Hrs____
| |_____	Hrs____
| |_____	Hrs____

Click or tap here to enter text.

Old Items:

Click or tap here to enter text.

Current Items:

Click or tap here to enter text.

New Items:

Click or tap here to enter text.

!!! URGENT ITEMS !!!

Click or tap here to enter text.

F-14	RUBBERFORM RECYCLED PRODUCTS, LLC	What:	Equipment Status
		Where:	
		Who:	
	Purpose:	Provide data collection point for Managers and Operators	

Machine Operation:

Click or tap here to enter text.

Machine Safety:

Click or tap here to enter text.

Machine Secondary Features:

Click or tap here to enter text.

Tooling:

Click or tap here to enter text.

Tooling Operation:

Click or tap here to enter text.

Operator Concerns:

Click or tap here to enter text.

Notes:

Click or tap here to enter text.

26.0

RUBBERFORM RECYCLED PRODUCTS, LLC	Date Created: 7/2/2025 Last Rev. Date: 7/5/2025 Revision Level: 0	**F-14** Yearly **Equipment Inspection**

Machine Name	
Machine ID	
Department	
Process	
Owner	
Approved By	

COMPONENTS NEEDED;

| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____

SPECIALIST NEEDED:

| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____
| |_____$_____

Click or tap here to enter text.

Old Items:

Click or tap here to enter text.

Current Items:

Click or tap here to enter text.

New Items

Click or tap here to enter text.

!!! URGENT ITEMS !!!

Click or tap here to enter text.

F-16	RUBBERFORM RECYCLED PRODUCTS, LLC	What:	Work Order Request Form
		Where:	All Departments
		Who:	Admin, Managers
	File Location	C:\Users\jesse\Documents\RUBBERFORM\QMS\CURRENT MASTER QMS\MASTER\Forms\F-16 Work Order Request.docx	

Project Name	
What	
Who	
Where	
Why	
Lead Person	
Department	
Priority	
Vendors	

Date Initiated:	
Initiated By:	
Given To:	
Need Completed By:	
Or What?	
Not to Exceed $$$	
Labor Resources in Hrs	
Approved By:	
Signature:	

JOB DESCRIPTION: Please explain what you need done.

REQUIREMENTS: Please explain the end goal, specific requirements and detailed expectations.

VARIABLES: Please explain things that are unknown or details that will / can change.

OTHER NOTES:

C:\Users\jesse\Documents\RUBBERFORM\QMS\CURRENT MASTER QMS\MASTER\Forms\F-16 Work Order Request.docx

Additional Details:

#	Qty	Part	Vendor	Price	Lead Time	Purchased	PO #	Due Date
1								
2								
3								
4								
5								
6								
7								
8								
9								

☐ **Introduction to the Quality Manual**

- Explained what the Quality Manual is and why it matters
- Shared how it supports everyone's job (not just office staff)
- Reinforced that it's a tool — not a burden

☐ **Overview of Key Sections**

- Quality Policy and company commitment to quality
- Roles and responsibilities
- Document control basics (where to find current procedures)
- Non-conformance reporting (how and why to report problems)
- Corrective action process overview
- Work instruction connection to the Quality Manual

☐ **Role-Specific Application**

- Reviewed relevant procedures or work instructions
- Demonstrated how the Quality Manual supports day-to-day tasks
- Reinforced expectations for following documented processes
- Emphasized the importance of accurate records and checklists

---

☐ **Communication & Continuous Improvement**

- Encouraged asking questions or suggesting improvements
- Shared how changes to the manual are handled (Document Change Requests)
- Explained who to go to for help or clarification

☐ **Follow-Up**

- Employee signed acknowledgment of Quality Manual introduction
- Follow-up review date scheduled (within 2–4 weeks)
- Notes/feedback from initial discussion recorded

Employee Signature: _____ **Date:** _____

Trainer/Supervisor Signature: _____ **Date:** _____

26.0

RubberForm – Weekly Maintenance Meeting Agenda

Meeting Title: Weekly Maintenance Review
Facilitator: Maintenance Supervisor
Attendees: Maintenance Techs, Production Rep, QA (optional)
Frequency: Weekly
Duration: 30–45 minutes

1. **Agenda:**
2. **Open Work Orders Review**
   - Status of current issues (machines, tools, facility)
3. **Preventive Maintenance Schedule**
   - What's due this week
   - Any missed or rescheduled tasks
4. **Critical Equipment Review**
   - Top risk machines for breakdown
   - Repeat problems and root cause follow-up
5. **Parts and Supplies Inventory**
   - Orders pending
   - Critical spares needed
6. **Safety & Compliance**
   - Lockout/tagout issues
   - Emergency repair protocols
   - OSHA-related concerns
7. **Collaboration with Production**
   - Planned downtime coordination
   - Feedback from operators on machine behavior
8. **Continuous Improvement / 5S**
   - Facility organization, reliability improvements
   - Suggestions from team
9. **Assign New Action Items & Owners**

Meeting Title: Weekly Production Operations Review
Facilitator: Production Manager
Attendees: Shift Leads, Operators, QA, Logistics, Maintenance
Frequency: Weekly
Duration: 30–45 minutes

1. **Agenda:**
1. **Safety Moment / Quick Safety Tip**
    - Any recent incidents or near misses
2. **Review of Production Performance**
    - Output vs. plan
    - Scrap/rework rates
    - Production metrics (if applicable)
3. **Schedule & Planning**
    - Review of weekly production schedule
    - Upcoming jobs or customer deadlines
    - Material availability
4. **Quality Issues**
    - In-process defects
    - Customer returns
    - Feedback from quality inspections
5. **Workforce Updates**
    - Staffing levels, training needs
    - Attendance or team support notes
6. **Maintenance Status**
    - Equipment availability
    - Open maintenance tickets impacting production
7. **Opportunities for Improvement**
    - Suggestions from operators
    - Ideas for better workflow or efficiency
8. **Assign New Action Items & Owners**

Meeting Title: Engineering Weekly Alignment
Facilitator: Engineering Manager
Attendees: Production, Maintenance, Sales
Frequency: Weekly
Duration: <30 minutes

1 Agenda:

1. **Review of Previous Action Items**
   - Status of outstanding tasks
2. **New Project Updates**
   - Engineering changes (ECNs/ECRs)
3. **Document & Drawing Control**
   - Review of recent document revisions (DCRs)
   - CAD file and BOM accuracy checks
4. **CAPA & Nonconformance Follow-Up**
   - Engineering role in investigations or root cause analysis
   - Preventive actions and solution validation
5. **Continuous Improvement Initiatives**
   - Lean projects, design for manufacturability (DFM), or automation ideas
6. **New Business / Open Discussion**
   - Questions, concerns, or improvement suggestions
7. **Assign New Action Items & Owners**

26.0

1. **Purpose**
   1.1. Ensure that compression molds achieve the required surface finish and dimensional accuracy to produce defect-free parts and extend mold life.
2. **Scope**
   2.1. Applies to all compression molds used in production, including single-cavity and multi-cavity steel or aluminum molds.
   2.2. Excludes initial mold fabrication (machining) and chemical etching processes.
3. **References**
   3.1. ISO 9001:2015 Clause 8.5 "Production and service provision"
   3.2. Mold manufacturer's polishing guidelines
   3.3. ANSI B46.1 "Surface Roughness"
   3.4. Equipment Safety Manual (Polishing Machine)
   3.5. GMP § 6.2 "Equipment Maintenance"
4. **Definitions**
   4.1. **Cavity Surface:** The internal surface of the mold that directly contacts the part.
   4.2. **Roughness (Ra):** Arithmetic average of surface peaks and valleys; target typically 16 µin (0.4 µm) or better.
   4.3. **Polishing Media:** Abrasive compounds (e.g., diamond paste, aluminum oxide) used to smooth surfaces.
5. **Responsibilities**
   5.1. **Mold Technician** – Execute polishing steps and document parameters; ensure safety protocols are followed.
   5.2. **Quality Inspector** – Verify surface finish and dimensional tolerances post-polish; approve mold for production or return to technician for rework.
   5.3. **Maintenance Manager** – Maintain polishing equipment and ensure abrasive inventories.
6. **Procedure**
   6.1. **Preparation**
   6.1.1. Review the mold drawing and surface finish requirements.
   6.1.2. Inspect the mold for damage, corrosion, or residue—record any defects.
   6.1.3. Clean the mold with a mild solvent and dry thoroughly.
   6.1.4. Mount the mold securely in the polishing fixture or machine.
   6.2. **Coarse Polishing**
   6.2.1. Select coarse abrasive compound (e.g., 320 grit aluminum oxide).
   6.2.2. Apply compound to the polishing tool (rotary or hand pad).
   6.2.3. Using light, consistent pressure, polish all cavity surfaces in sweeping motions.
   6.2.4. Rinse the mold to remove swarf; inspect under magnification for deep scratches or tool marks.
   6.3. **Intermediate Polishing**
   6.3.1. Switch to medium abrasive (e.g., 600 grit).
   6.3.2. Repeat polishing strokes, focusing on removing scratch patterns from the previous step.
   6.3.3. Clean and dry mold; measure surface roughness at multiple points.

6.4. **Fine Polishing**
6.4.1. Use fine/diamond paste (1–3 µm particles).
6.4.2. Polish until a uniform, mirror-like finish is achieved.
6.4.3. Wipe down with lint-free cloth and a solvent approved for mold cleaning.
6.5. **Final Inspection**
6.5.1. Measure surface roughness (Ra) with a profilometer—verify against specification.
6.5.2. Check critical dimensions with calipers or a CMM as needed.
6.5.3. Record results on the Polishing Inspection Report.
6.5.4. If any parameter is out of tolerance, return mold to the technician for targeted rework.
6.6. **Post-Polish Cleaning & Storage**
6.6.1. Perform ultrasonic cleaning if available, or wipe with recommended solvent.
6.6.2. Apply a thin coat of rust preventive if the mold will be stored.
6.6.3. Label and store the mold in the designated area.

7. **Supporting Documents**
   7.1. Polishing Work Instruction (detailed steps and machine settings)
   7.2. Polishing Media Log (tracks batch, grit size, and usage)
   7.3. Polishing Inspection Report (finish measurements and dimensional check)
   7.4. Equipment Maintenance Checklist (daily/weekly checks for polishing machine)
8. **Recordkeeping**
   8.1. Polishing Inspection Reports: retain for 2 years in `04_Records/Inspections/Polishing`
   8.2. Media Logs: keep abrasive usage records for 1 year in `05_Forms/Media_Logs`
   8.3. Equipment Maintenance Logs: store per Maintenance procedure retention (see Procedure 15)
   8.4. All records must be legible, signed, and date-stamped.

9. Revision History

1. Any modifications must be documented with a revision history, including changes made and the reason for the update.
2. Revised documents require review and approval before implementation.
3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release-Document Control	J Horanburg	J Probst	10/1/2025
A				
B				

Hydraulic Cylinder Replacement Procedure

1. **Purpose**
   1.1 Ensure safe and effective removal and installation of hydraulic cylinders to restore system performance and prevent leaks or failures.
2. **Scope**
   2.1 Applies to all stationary and mobile equipment using hydraulic cylinders.
   2.2 Excludes routine cylinder maintenance (see Preventive Maintenance Procedure).
3. **References**
   3.1 ISO 9001: Clause 8.5.1 "Control of production and service provision"
   3.2 OSHA 29 CFR 1910.147 "Control of Hazardous Energy (Lockout/Tagout)"
   3.3 Equipment manufacturer's service manual
   3.4 Hydraulic fluid Safety Data Sheet (SDS)
4. **Definitions**
   4.1 **Hydraulic Cylinder** – Actuator converting hydraulic pressure into linear force and motion.
   4.2 **Rod End** – The piston rod that extends from the cylinder barrel.
   4.3 **Ports** – Inlet and outlet fittings for hydraulic fluid.
   4.4 **LOTO** – Lockout/Tagout, a safety procedure to isolate energy sources.
5. **Responsibilities**
   5.1 **Maintenance Technician** – Execute replacement steps, complete forms, perform functional test.
   5.2 **Production Supervisor** – Schedule downtime, verify system return to service.
   5.3 **Safety Officer** – Approve LOTO permit and monitor compliance.
6. **Procedure**
   6.1 **Preparation**
   6.1.1 Obtain new cylinder that matches part number and specifications.
   6.1.2 Review manufacturer's manual for torque values and seal orientation.
   6.1.3 Ensure work area is clean and free of debris.
   6.2 **Energy Isolation & Fluid Drain**
   6.2.1 Follow LOTO Procedure to de-energize hydraulic system and lock out power.
   NOTE: De-energizing includes relieving pressure within the system and ensuring all moving components are in their ZERO energy position/state.
   6.2.2 Place catch pan under cylinder ports.
   6.2.3 Slowly loosen port fittings to drain residual fluid; cap ports and collect fluid in approved container.
   6.3 **Cylinder Removal**
   6.3.1 Remove mounting hardware (pins, bolts) supporting cylinder barrel and rod end.
   6.3.2 Carefully withdraw cylinder from its mounts, supporting its weight to prevent damage.

6.3.3 Inspect rod and barrel for scoring or corrosion; log any findings on Inspection Checklist.

6.4 **Installation of New Cylinder**

6.4.1 Position new cylinder in mounts; insert pins/bolts and torque to manufacturer's specification.

6.4.2 Reconnect hydraulic hoses to ports, using new sealing washers if required; torque fittings per manual.

6.5 **Refill & Bleed System**

6.5.1 Remove caps; refill system reservoir with specified hydraulic fluid to recommended level.

6.5.2 Bleed air from cylinder by cycling it slowly through full stroke until smooth operation is achieved.

6.6 **Functional Test**

6.6.1 Remove LOTO devices following safety procedure.

6.6.2 Operate equipment through several cycles under no-load conditions; check for leaks and proper travel.

6.6.3 Pass load test under normal working pressure; verify no drift or hesitation.

6.7 **Return to Service**

6.7.1 Clean any spilled fluid and restore work area.

6.7.2 Inform Production Supervisor that equipment is ready.

7. **Supporting Documents**

7.1 Hydraulic Cylinder Replacement Work Order

7.2 Lockout/Tagout Permit

7.3 Cylinder Inspection Checklist

7.4 Fluid Collection & Disposal Log

8. **Recordkeeping**

8.1 Retain Work Order, LOTO permit, and Inspection Checklist for 3 years in `04_Records/Maintenance`.

8.2 File Fluid Disposal Log per environmental regulations for 2 years in `05_Forms/Environmental`.

8.3 Ensure all records are signed, dated, and legible.

9. Revision History

1. Any modifications must be documented with a revision history, including changes made and the reason for the update.
2. Revised documents require review and approval before implementation.
3. Old versions must be archived and marked as "Obsolete."

Revision	Reason	Revised by	Approved by	Date
Original	Initial Release	J Horanburg	J Probst	10/1/2025
A				
B				